**Task 2: Data Retrieval & Sorting Dashboard**

**Aim**

To create a **dashboard page** that displays student or employee records from the database with features such as **sorting by name or date**, **filtering by department**, and **displaying the count of students in each department**.

**Algorithm**

1. Start the program.
2. Connect the dashboard page to the database.
3. Retrieve all student or employee records using the SELECT query.
4. Display the retrieved records in a table format.
5. Provide sorting options for:
    - Name
    - Date
6. When the user selects sorting, rearrange the records accordingly.
7. Provide filtering option based on department.
8. When the user selects a department, show only matching records.
9. Count the number of students available in each department.
10. Display the department-wise count clearly on the dashboard.
11. Refresh the displayed records dynamically whenever sorting or filtering is applied.
12. Stop the program.

**Program :**

**Index.html:**

```
<!DOCTYPE html>
<html>
<head>
<title>Register</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
<h2>Register Student</h2>
<form id="form">
```

```
<input name="name" placeholder="Name" required>
<input name="email" placeholder="Email" required>
<input type="date" name="dob" required>
<select name="department">
<option>CSE</option>
<option>IT</option>
<option>ECE</option>
<option>EEE</option>
</select>
<input name="phone" placeholder="Phone">
<button>Submit</button>
</form>
<a href="dashboard.html">Go to Dashboard →</a>
</div>
<script>
document.getElementById("form").onsubmit = async (e) => {
  e.preventDefault();
  const data = Object.fromEntries(new FormData(e.target));
  await fetch("/addStudent", {
    method:"POST",
    headers:{"Content-Type":"application/json"},
    body:JSON.stringify(data)
  });
  alert("Student Added!");
};
</script>
</body>
</html>
```

**Dashboard.html:**

```
<!DOCTYPE html>
<html>
```

```
<head>
<title>Dashboard</title>
<link rel="stylesheet" href="style.css">
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="https://cdn.jsdelivr.net/npm/tsparticles@2/tsparticles.bundle.min.js"></script>
</head>
<body>
<div id="particles"></div>
<div class="container">
<h2>Dashboard</h2>
<input id="search" placeholder="Search" oninput="suggest()">
<ul id="suggestions"></ul>
<select id="sort">
<option value="">Sort</option>
<option value="name">Name</option>
<option value="dob">DOB</option>
</select>
<select id="dept">
<option value="">All</option>
<option>CSE</option>
<option>IT</option>
<option>ECE</option>
<option>EEE</option>
</select>
<button onclick="load()">Apply</button>
<table>
<thead>
<tr>
<th>Name</th><th>Email</th><th>DOB</th><th>Dept</th><th>Phone</th>
</tr>
</thead>
```

```
<tbody id="body"></tbody>
</table>
<div class="pagination">
<button onclick="prev()">Prev</button>
<span id="page">1</span>
<button onclick="next()">Next</button>
</div>
<canvas id="chart"></canvas>
</div>
<script>
let page = 1;
async function load() {
   const search = document.getElementById("search").value;
   const sort = document.getElementById("sort").value;
   const dept = document.getElementById("dept").value;
   const res = await
fetch(`/students?search=${search}&sortBy=${sort}&department=${dept}&page=${page}`);
   const data = await res.json();
   const body = document.getElementById("body");
   body.innerHTML = "";
   data.forEach(s=>{
      body.innerHTML += `<tr>
      <td>${s.name}</td>
      <td>${s.email}</td>
      <td>${s.dob}</td>
      <td>${s.department}</td>
      <td>${s.phone}</td>
      </tr>`;
   });
   document.getElementById("page").innerText = page;
}
```

```
function next(){ page++; load(); }
function prev(){ if(page>1) page--; load(); }
// Suggestions
async function suggest() {
   const val = document.getElementById("search").value;
   if(!val) return;
   const res = await fetch(`/students?search=${val}`);
   const data = await res.json();
   let html="";
   data.slice(0,5).forEach(s=>{
      html+=`<li onclick="pick('${s.name}')">${s.name}</li>
});
   document.getElementById("suggestions").innerHTML = html;
}
function pick(name){
   document.getElementById("search").value=name;
   document.getElementById("suggestions").innerHTML="";
   load();
}
// Chart
async function chart(){
   const res = await fetch("/departmentCount");
   const data = await res.json();
   new Chart(document.getElementById("chart"), {
      type:"bar",
      data:{
         labels:data.map(d=>d.department),
         datasets:[{data:data.map(d=>d.count)}]
      }
   });
}
```

```
// Particles
tsParticles.load("particles", {
  particles: { number:{value:40}, size:{value:3}, move:{speed:1} }
});
load();
chart();
</script>
</body>
</html>
```

**Style.css:**

```
@import
url('https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap');
* { margin:0; padding:0; box-sizing:border-box; }
body {
  font-family: Poppins;
  background: linear-gradient(-45deg,#0f2027,#2c5364,#3a1c71);
  background-size: 400% 400%;
  animation: bg 10s infinite;
  color: white;
}
@keyframes bg {
  0%{background-position:0%}
  50%{background-position:100%}
  100%{background-position:0%}
}
/* Particle layer */
#particles {
  position: fixed;
  width: 100%;
  height: 100%;
  z-index: -1;
```

```
}
/* Container */
.container {
    width: 90%;
    max-width: 900px;
    margin: 40px auto;
    padding: 20px;
    border-radius: 15px;
    background: rgba(255,255,255,0.1);
    backdrop-filter: blur(15px);
}
/* Inputs */
input, select {
    padding:10px;
    margin:5px;
    border-radius:8px;
    border:none;
}
/* Buttons */
button {
    padding:10px;
    border:none;
    border-radius:8px;
    background:#ff4b2b;align-items: ;
    color:white;
    cursor:pointer;
}
/* Table */
table {
    width:100%;
    margin-top:10px;
```

```
}
td, th {
    padding:8px;
    text-align:center;
}
tr:hover {
    background:rgba(255,255,255,0.2);
}
/* Suggestions */
#suggestions {
    background:rgba(0,0,0,0.6);
    border-radius:8px;
    list-style:none;
}
#suggestions li {
    padding:5px;
    cursor:pointer;
}
/* Pagination */
.pagination {
    text-align:center;
    margin-top:10px;
}
```

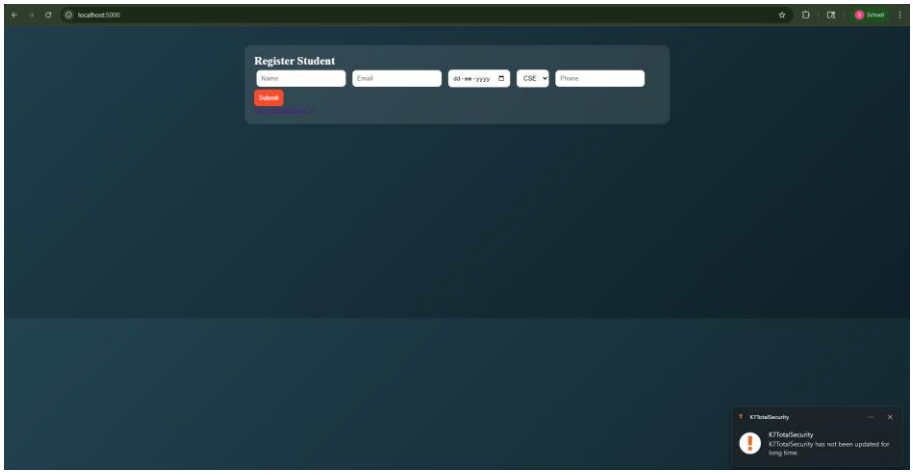
**Server.js:**

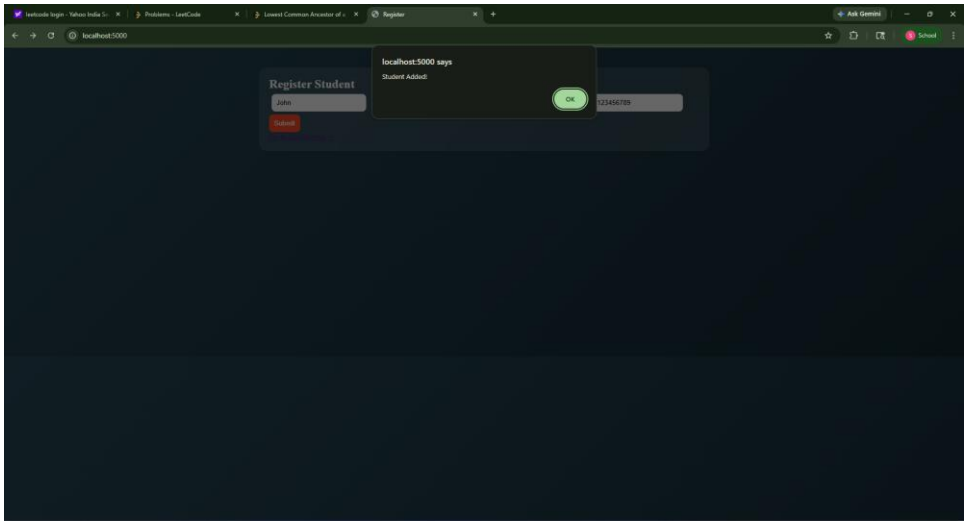
```
const express = require("express");
const mysql = require("mysql2");
const cors = require("cors");
const app = express();
app.use(cors());
app.use(express.json());
app.use(express.static("public"));
```

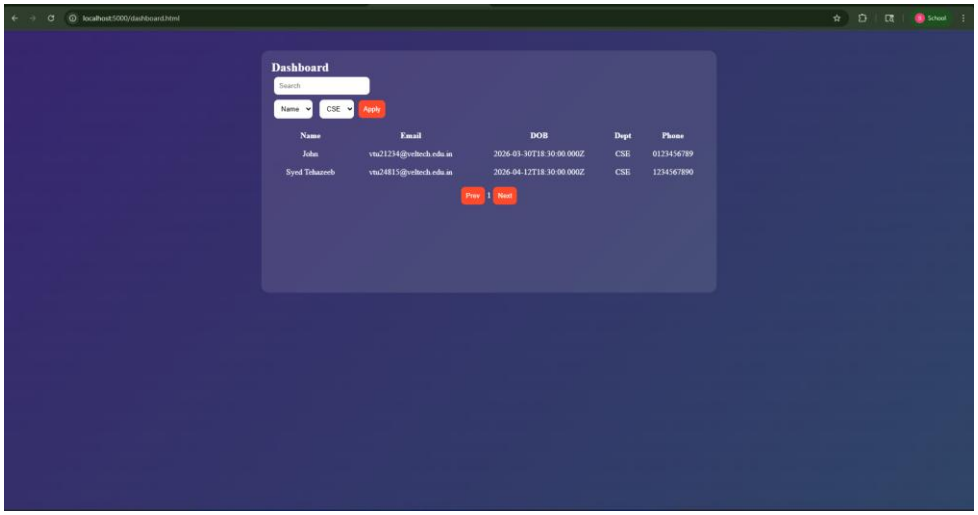
```
const db = mysql.createConnection({
  host: "localhost",
  user: "root",
  password: "tehazeeb@2005",
  database: "student_db"
});
// Add student
app.post("/addStudent", (req, res) => {
  const { name, email, dob, department, phone } = req.body;
  const sql = "INSERT INTO students (name,email,dob,department,phone) VALUES (?,?,?,?,?)";
  db.query(sql, [name, email, dob, department, phone], (err) => {
    if (err) return res.send(err);
    res.send("Added");
  });
});
// Get students (search + filter + sort + pagination)
app.get("/students", (req, res) => {
  let { search, sortBy, department, page = 1, limit = 5 } = req.query;
  page = parseInt(page);
  limit = parseInt(limit);
  const offset = (page - 1) * limit;
  let sql = "SELECT * FROM students WHERE 1=1";
  let values = [];
  if (search) {
    sql += " AND name LIKE ?";
    values.push(`%${search}%`);
  }
  if (department) {
    sql += " AND department = ?";
    values.push(department);
  }
```

```
    if (sortBy === "name") sql += " ORDER BY name ASC";
    if (sortBy === "dob") sql += " ORDER BY dob ASC";
    sql += " LIMIT ? OFFSET ?";
    values.push(limit, offset);
    db.query(sql, values, (err, result) => {
        if (err) return res.send(err);
        res.json(result);
    });
});
// Department count
app.get("/departmentCount", (req, res) => {
    db.query(
        "SELECT department, COUNT(*) as count FROM students GROUP BY department",
        (err, result) => {
            if (err) return res.send(err);
            res.json(result);
        }
    );
});
app.listen(5000, () => console.log("Server running on port 5000"));
```

Outputs:






Result:

To create a **dashboard page** that displays student or employee records from the database with features such as **sorting by name or date**, **filtering by department**, and **displaying the count of students in each department** is executed successfully.